

HORTA AP, Portugal

WMO: 085050

Lat: 38.5211N

Lon: 28.7139W

Elev: 41

StdP: 100.83

Time Zone: -1.00 (AZR)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.0	10.9	3.9	5.0	11.5	4.9	5.4	12.1	15.2	14.3	13.9	14.5	4.9	0	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	26.0	22.1	25.2	21.6	24.8	21.4	23.2	24.6	22.7	24.2	22.3	23.8	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.7	17.5	24.1	22.2	16.9	23.7	21.8	16.6	23.4	68.6	24.7	67.0	24.3	65.4	23.8	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.4	10.1	8.0	27.5	1.0	1.0	7.3	28.3	6.6	28.9	6.1	29.4	5.3	30.2		
(5)			5.8	24.5	1.0	0.7	5.1	25.0	4.5	25.4	4.0	25.8	3.2	26.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	15.1	14.9	15.0	15.5	16.7	19.0	21.5	22.7	21.8	19.5	17.2	15.9	(6)
(7)		DBStd	3.17	1.71	1.61	1.55	1.38	1.37	1.31	1.34	1.26	1.38	1.74	1.86	1.61	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	576	101	97	104	85	53	8	0	0	0	8	43	76	(9)
(10)		CDD10.0	2891	158	136	154	165	207	271	358	392	355	295	217	183	(10)
(11)		CDD18.3	424	0	0	0	0	2	28	99	134	105	45	9	1	(11)
(12)		CDH23.3	583	0	0	0	0	0	2	121	318	133	8	1	0	(12)
(13)		CDH26.7	6	0	0	0	0	0	0	1	4	1	0	0	0	(13)
(14)	Wind	WSAvg	4.8	5.8	5.7	5.7	4.9	4.5	3.9	3.2	3.5	4.1	4.7	5.2	5.8	(14)
(15)	Precipitation	PrecAvg	1045	105	88	107	70	80	58	37	50	85	124	108	134	(15)
(16)		PrecMax	1608	198	180	254	146	257	126	155	209	152	284	234	296	(16)
(17)		PrecMin	702	9	12	28	15	1	11	6	4	27	22	14	19	(17)
(18)		PrecStd	197	44	48	62	41	65	32	28	40	42	66	60	65	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	18.8	19.1	20.1	21.1	23.5	26.2	27.1	26.2	24.2	22.9	19.2	(19)
(20)			MCWB	16.7	16.4	15.9	16.8	18.2	20.2	22.3	22.2	22.4	21.1	19.2	17.4	(20)
(21)		2%	DB	18.1	18.0	18.2	19.1	20.2	22.4	25.2	26.1	25.2	23.2	21.2	19.0	(21)
(22)			MCWB	16.4	16.0	15.6	16.5	17.6	19.5	21.5	22.1	21.7	20.7	19.2	17.4	(22)
(23)		5%	DB	17.9	17.3	17.7	18.2	19.7	21.9	24.6	25.7	24.8	22.8	20.5	18.4	(23)
(24)			MCWB	16.3	15.6	15.3	15.9	17.2	19.2	21.1	21.9	21.4	20.4	18.7	16.9	(24)
(25)		10%	DB	17.2	17.1	17.1	17.8	19.0	21.1	23.9	25.0	24.1	22.1	19.9	18.1	(25)
(26)			MCWB	15.9	15.5	15.1	15.6	16.8	18.6	20.7	21.7	20.8	19.9	18.2	16.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	17.2	17.2	18.1	19.2	21.5	23.7	24.0	23.4	22.5	20.8	18.8	(27)
(28)			MCDWB	18.3	17.9	18.0	19.2	20.1	22.6	25.0	25.5	24.9	23.3	21.5	19.0	(28)
(29)		2%	WB	17.2	16.7	16.6	17.3	18.6	20.6	22.9	23.3	22.8	21.9	20.1	18.1	(29)
(30)			MCDWB	17.8	17.4	17.5	18.3	19.5	21.7	24.2	24.9	24.2	22.7	20.7	18.4	(30)
(31)		5%	WB	16.9	16.3	16.1	16.7	18.1	20.0	22.3	22.9	22.4	21.3	19.4	17.6	(31)
(32)			MCDWB	17.4	17.1	17.1	17.8	19.0	21.1	23.7	24.5	23.8	22.3	20.1	18.2	(32)
(33)		10%	WB	16.5	15.9	15.7	16.2	17.6	19.6	21.7	22.5	21.9	20.6	18.7	17.2	(33)
(34)			MCDWB	17.2	16.8	16.8	17.3	18.6	20.7	23.2	24.2	23.4	21.8	19.6	18.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.3	3.6	3.7	3.8	3.9	4.0	4.7	4.6	4.4	4.0	3.7	3.2	(35)
(36)			MCDBR	3.1	3.5	3.8	4.1	4.3	4.5	5.1	5.0	4.7	4.1	3.9	3.0	(36)
(37)			MCWBR	3.3	3.3	3.1	3.5	3.3	3.2	3.4	3.3	3.3	3.3	3.4	3.1	(37)
(38)		5% WB	MCDWR	2.6	3.1	3.1	3.6	3.3	3.6	4.1	4.2	3.8	3.3	3.5	2.4	(38)
(39)			MCWBR	3.2	3.3	2.9	3.6	3.1	3.1	3.3	3.4	3.2	3.4	3.6	3.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.355	0.363	0.392	0.385	0.380	0.379	0.375	0.369	0.367	0.369	0.355	0.356	(40)	
(41)		taud	2.502	2.483	2.383	2.429	2.473	2.509	2.533	2.571	2.576	2.552	2.536	2.478	(41)	
(42)		Ebn at Noon	818	861	866	893	901	898	899	898	883	844	812	785	(42)	
(43)		Edh at Noon	80	92	111	112	110	106	102	96	91	85	77	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.01	2.86	3.98	5.17	5.96	6.35	6.54	6.17	4.83	3.37	2.26	1.73	(44)	
(45)		RadStd	0.16	0.15	0.28	0.46	0.46	0.42	0.35	0.25	0.24	0.22	0.13	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	+1.02	N/A	+0.35	+0.52	N/A	N/A	N/A	N/A

Nomenclature: See separate page